# PHANTOM PHOENIX: Image-Detector Accuracy Review

Independent ML research review, 16 Oct 2026. Sources were browsed (model cards, repos, arXiv, dataset cards); your artifacts were not run. Anything about your repo or data is a hypothesis until reproduced.

## 1. Executive diagnosis

- **No root cause is established.** The evidence supports a combination of three things, none yet tested on your system: test generators newer than the detector's training data, a 0.5 operating point, and a 45-image test set whose labels are confounded with data source.
- **Check model integrity before any science.** The HF repo you pin is a third-party port. Its maintainers say the weights, config and preprocessor were wrong, and fixed them in July 2026.
- **Your ranking signal is partly intact.** AUC of 0.82-0.90 with 100% fake precision and 25% recall points to scores shifted low for fakes, not random output.
- **The 85% target cannot be assessed with the current data.** The 95% interval on 51.1% accuracy (23/45) is roughly 37-65%. Fake recall of 7/28 is roughly 13-43%. The pilot (6/25) gives roughly 11-43%. AUC 0.82 on n=45 has an approximate +/-0.12 interval.
- **The strongest literature signal is about data.** On modern generators, detectors trained on older generators show the same real-class bias you see. Training on modern generator data fixed much of it in one preprint.

## 2. Confirmed facts vs hypotheses

**Confirmed from sources:**

- The upstream model is Park & Owens, CVPR 2025. Official checkpoints are `OwensLab/commfor-model-384` and `-224` (MIT), alongside the Community Forensics, Small and Eval datasets.
- The `buildborderless` repo is a port. Its maintainers state the safetensors weights came from different weights than the training checkpoint, the config carried ViT-Base values (12 heads, 3072 intermediate) instead of ViT-Small (6 heads, 1536), and the preprocessor squashed non-square images. The fix requires transformers >= 5.4.0.

- OpenFake's license differs across revisions of its card (CC-BY-NC-4.0, CC-BY-NC-SA-4.0, CC-BY-SA-4.0). The current card says the test split pairs held-out generators with real images from sources distinct from training reals.
- Your supplied arithmetic is internally consistent: the confusion matrices reproduce the stated precision, recall, F1 and accuracy.

**Unverified:**

- Whether pinned revision `ac6ee457...` is pre-fix or post-fix. Your 440/384 preprocessing matches the post-fix description, but that is not proof.
- Your actual preprocessing, normalization and label direction.
- Dataset composition, resolution and JPEG distribution, and any near-duplicates.
- That Community Forensics fails on FLUX.2/Z-Image specifically. No benchmark found evaluated it on them.

## 3. Ranked causes of low recall

1. **Port or preprocessing integrity (test first).** Plausibility medium: AUC well above 0.5 argues against total breakage, not partial degradation. Under a day to test. Two third-party cards disagree on normalization (CLIP vs ImageNet statistics), and one says the resampling filter matters.
2. **Unseen-generator shift.** Plausibility high: z-image-turbo and flux.2-klein postdate the 2024 paper. Analogy only: official detectors in DailyBench (Jul 2026 preprint, not peer-reviewed) show high real accuracy and near-zero fake accuracy on FLUX.2 and Z-Image (UniFD 0.50% fake accuracy on FLUX.2). Test with per-generator scores and a probe trained on held-out generators.
3. **Operating point and calibration.** Plausible given AUC 0.82-0.90. Cheap to test, but with 24 validation images the threshold cannot be chosen reliably.
4. **Source-label confound.** Reals are ImageNet and DOCCI; fakes are generator outputs. Resolution, JPEG history and style may separate classes. This inflates specificity (25/25 and 16/17 reals correct) and makes recall hard to interpret. Test with a format-only classifier (width, height, file size, JPEG quantization tables, EXIF).
5. **Small-sample variance.** Wide intervals, though two runs at 24-25% recall agree.
6. **Label noise.** Low prior. Hand-check 50 low-scoring "fakes" and 50 high-scoring "reals".
7. **Evaluation bugs.** Low: precision of 100% and AUC above 0.5 imply the label direction is right. Confirm anyway.

8. **Leakage.** Cannot explain low accuracy, but affects trust in any trained result. You have a train split despite a frozen model, so check it.

## 4. Candidate models

Size estimates for the 4 GB GPU are mine, not measured.

| Candidate | Source / license | Evidence | Fit |
|---|---|---|---|
| Official `commfor-model-384` | OwensLab HF, MIT | Control for the port. No modern-generator benchmark found. | ViT-S, ~22M params, fits easily |
| B-Free | CVPR 2025, arXiv 2412.17671 | Code and data licensed for informational and nonprofit use only. Official checkpoint scored 52.7% average on DailyBench FakeBench (5% fake accuracy on FLUX.2) but 88.6% balanced accuracy on in-the-wild sets | Inference likely fits; verify checkpoint size |
| NPR | CVPR 2024 | Official checkpoint ~60% real and 60-90% fake accuracy on modern generators in DailyBench: noisy but non-zero fake recall | Small CNN, CPU feasible |

| Candidate | Source / license | Evidence | Fit |
|---|---|---|---|
| UniversalFakeDetect (CLIP ViT-L/14 + linear head) | CVPR 2023 | Near-zero fake accuracy on modern generators with official weights | Skip as detector; the linear-probe method is reusable |
| CNNDetection (ProGAN ResNet-50) | CVPR 2020 | Negative control only | Trivial |
| DINOv3 / CLIP / PE frozen features + head | arXiv 2508.10104 and others | arXiv 2509.12995 reports modern foundation models generalize well in the wild. DailyBench's baseline used DINOv3-L with LoRA | ViT-L fp16 weights ~0.6 GB: extraction should fit; fine-tuning does not. DINOv3 license unverified |
| Third-party Community Forensics fine-tunes (Thermostatic, angelhd25, Vontra) | HF, MIT-derived | Self-reported on their own data; one states its calibration set was also used for checkpoint selection | External check only; unknown training data may overlap yours |

PROBE, DDA and DINO-Det: checkpoint availability and licenses not verified. DINO-Det needs a 7B encoder and is infeasible here.

Do not assume generalization from older GAN/diffusion data transfers. AIGIBench finds many detectors misclassify nearly all samples from newer text-to-image models and in-the-wild sources, and that test-time resizing mainly raises real-image accuracy, not fake.

## 5. Recommended experiments, in priority order

Do not run E6 or any ensemble before E0, E1 and E5.

| ID | Hypothesis | Input | Metrics | Decision rule |
|---|---|---|---|---|
| E0 | Pinned revision or pipeline differs from official | Same 200 images through (a) production path, (b) official OwensLab ckpt via its own loader, (c) current post-fix HF | Logit diff, AUC, confusion at 0.5 | If AUC differs >= 0.05 or logits disagree materially, replace the model with official before anything else |
| E1 | Dataset confounds explain metrics | Manifests, EXIF, JPEG tables, resolution | Format-only classifier AUC, per-source and per-generator score distributions | If format-only AUC is high, rebuild the test set with matched formats |
| E2 | Preprocessing hurts | Dev set; resize 440+crop 384 vs native crop vs full-image resize vs multi-crop mean | AUC, recall at fixed FPR | Adopt only if the gain exceeds the bootstrap interval on dev, with no test peeking |
| E3 | Threshold/calibration | Cross-validated scores on train+val (frozen model, so all 155 usable) | Recall at target FPR, ECE | Adopt only if the interval clears baseline; otherwise report as unresolved |
| E4 | Another detector generalizes better | Official B-Free, NPR, official CF | Same, per generator | Advance a candidate only if it beats the control and its errors differ |

| ID | Hypothesis | Input | Metrics | Decision rule |
|---|---|---|---|---|
| E5 | Frozen features separate modern fakes | >= 2-5k images, >= 3 generators; leave-one-generator-out | AUC, recall at fixed FPR on held-out generator | If held-out AUC is poor, features or data are inadequate; do not fine-tune |
| E6 | Fine-tuning helps | Colab T4; >= 10k images, >= 6 generators; fake-preferring batches | Same, plus robustness | Only if E5 passes. DailyBench's ablation favored about 5:3 fake:real batches |
| E7 | Robustness | JPEG q50/75/90, resize, screenshot | Recall/FPR change | Defines the abstention domain |

## 6. Data requirements and strategy

- 200 images is enough only for diagnostics: a logistic head on frozen features as a sanity check and score-distribution audits. It is not enough for fine-tuning, threshold selection or per-generator claims.
- Because source equals label, any trained model will learn source cues.
- Sizing guidance (my arithmetic): demonstrating fake recall around 80% with a +/-7-point interval needs about 125 fakes per reported generator family. Estimating a ~1% false-positive rate needs about 300 reals with zero errors.
- Suggested locked test set: >= 300 reals and >= 300 fakes, >= 100 per priority generator, plus a held-out generator never used in development.
- Reals should be format-matched to fakes. Include phone photos, social-media recompressed images and screenshots.
- Check licenses per source. OpenFake reals come from LAION, Pexels and others, so the dataset license does not cover underlying content. Community Forensics-Small is documented as paired with redistributable-license reals. Use OpenFake for research only; pin the dataset revision and the license text at that revision.

## 7. Compute requirements (RTX 3050 Laptop, ~4 GB)

- Inference of ViT-S, NPR and B-Free should fit. Run a measured latency/VRAM benchmark (fp32 vs fp16, batch 1 and 8) and record it; these were not measured here.
- Frozen ViT-L feature extraction should fit in fp16; verify.
- Fine-tuning: use Colab GPU once E5 passes. Keep linear heads on the laptop. Colab availability and tier are not guaranteed.

## 8. Training vs inference recommendations

- Production stays inference-only until E0-E5 finish.
- Escalation path: (1) official checkpoint, replacing the port if E0 shows a difference; (2) re-thresholded or calibrated baseline; (3) frozen-feature head; (4) partial fine-tune of ViT-S.
- Full fine-tuning or a compact CNN on 200 images is not justified. A frequency/residual branch needs the data first.

## 9. Evaluation and leakage-prevention protocol

- Splits are group-aware by source, generator family and prompt/real-pair.
- Near-duplicates: compute perceptual hash plus embedding similarity across all splits; set the threshold by inspecting borderline pairs, then freeze it.
- Lock the final test set. Touch it once per finalist, ideally once overall.
- Report: accuracy; per-class precision/recall/F1; confusion matrix; ROC-AUC and PR-AUC; FPR and FNR; per-generator results where n supports them, with Wilson/bootstrap intervals; latency and memory; ECE only if probability claims are made.
- Accuracy alone is insufficient: it hides class asymmetry and depends on class mix (your test is 62% fake), and false accusation of real media and missed fakes carry different costs.
- Cross-generator tests: hold out whole generator families, including an unseen-generator split.
- Run both native-format and format-normalized evaluations.

## 10. Threshold and calibration recommendations

- Set operating points by a target real-image FPR chosen from the cost of false accusation, not a 0.5 default.
- Temperature or Platt scaling is feasible at hundreds of samples; isotonic needs on the order of a thousand.
- Calibration holds only for the domain it was fit on. Output score bands, not probabilities, outside the validated generator/format domain.
- With the current 24-image validation set, do not adopt any tuned threshold.

## 11. Ensemble decision

Not now. Reconsider only if two detectors each show validation AUC clearly above chance, calibrate to a common scale, and make measurably different errors (each catches a meaningful share of the other's misses).

- Use calibrated logit averaging first; stacking needs hundreds per class.
- Never treat a failed or unavailable model as a zero fake score. Drop it from the average and mark the result "reduced evidence".
- Disagreement across the decision boundary yields "inconclusive", not an average.
- Weights come from cross-validation, not hand-picking.

## 12. Risks, limitations and fallbacks

- The generator landscape moves faster than checkpoints; expect decay.
- Local edits and inpainting evade most detectors: DailyBench reports most near 50% on localized-edit sets.
- Compression and screenshots degrade artifact-based cues.
- Fallback if no detector meets the target FPR and recall on your locked test: ship the detector as one evidence line with abstention, and rely on provenance and metadata for the rest.
- Report honestly: "not reached, with intervals" beats a tuned 85%.

## 13. First implementation sprint (E0, E1 and harness)

Acceptance criteria:

- A single eval script writes JSON with model revision hash, weight hash, preprocessing config, library versions, dataset manifest hash and seed.
- It reproduces your reported confusion matrices or documents each discrepancy.
- The manifest records source, generator, license, resolution, format, sha256 and near-duplicate cluster per image.
- E0 report: parity between production, official and post-fix paths, with the decision recorded.
- E1 report: format-only classifier result, per-source and per-generator score distributions, label audit of 100 images.
- No test-set metric is used for any decision.

## 14. Product reporting policy

- Keep separate: AI-generation score, manipulation indicators, generator-family attribution, metadata/provenance, explainability, and verdict/risk.
- States: "AI-generation indicators detected", "No AI-generation evidence detected" (never "authentic"), "Inconclusive".
- Each report names the model, version, threshold, calibration domain, supported-generator list and abstention reason.
- Never claim an exact generator checkpoint, seed, prompt or human creator from pixels, and never infer intent.
- Generator-family attribution needs its own validated model; do not derive it from the binary score.

## 15. Sources

- Port fix: https://huggingface.co/buildborderless/CommunityForensics-DeepfakeDet-ViT/discussions/5
- Official checkpoints: https://huggingface.co/OwensLab
- Repo: https://github.com/JeongsooP/Community-Forensics; paper: arXiv 2411.04125
- OpenFake: https://huggingface.co/datasets/ComplexDataLab/OpenFake
- B-Free: arXiv 2412.17671; license: https://www.grip.unina.it/download/prog/B-Free/training_data/README.md
- DailyBench: arXiv 2607.24016 (preprint)
- AIGIBench: arXiv 2505.12335

- Dual Data Alignment: arXiv 2505.14359
- Foundation models in the wild: arXiv 2509.12995

## Instructions for OpenCode

The repo has not been inspected, so inspect before changing anything.

1. Locate the detector wrapper, preprocessing, threshold constants and any score-to-verdict logic. Report the actual resize filter, crop, normalization, dtype, attention mode and label direction.
2. Add a standalone evaluation module that does not touch production code. It must accept a manifest and a model spec and write the JSON in section 13.
3. Write a manifest builder for the 200 images: sha256, perceptual hash, source, generator, license, resolution, format, JPEG quantization tables and EXIF flag.
4. Implement E0: run the three model paths and compare logits. Do not change the pinned revision yet; output a parity report.
5. Implement E1: format-only classifier, per-source and per-generator score histograms, and a 100-image audit sheet.
6. Add group-aware splitting with near-duplicate clustering and a locked test list that stores only image hashes.
7. Add Wilson and bootstrap interval helpers and wire the metrics in section 9.
8. Open a decision record per experiment (hypothesis, result, decision). Do not change production thresholds or models without a recorded decision.